

Aditya Kaleeswar

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EDUCATION

Vellore Institute of Technology (VIT)

B.Tech in Computer Science and Engineering | CGPA: 8.41

Vellore, India
Sep 2022 - Jul 2026

EXPERIENCE

University of Texas Rio Grande Valley - AIM Lab

Research Intern

Remote
Jan 2026 - Jul 2026

ECC: [Preprint](#) | [GitHub](#) PINN: [Preprint](#) | [GitHub](#)

- Developed an uncertainty-aware ML model to predict tensile strength and ductility of engineered cementitious composites (ECC) from mix design using CatBoost quantile regression and conformal calibration; delivered 80.4%/82.6% empirical coverage at nominal 80% and R^2 of 0.75/0.61 despite high experimental variance.
- Designed a 37-feature representation from 18 raw mix variables, composition ratios, and micromechanical features; trained and calibrated on 620 specimens across 276 unique mixes using GroupKFold to prevent replicate leakage.
- Implemented constrained inverse mix-design optimization with NSGA-II to identify the lowest-cost lab-realizable mix for target strength-ductility specifications; rejected unsupported or infeasible targets and surfaced designs with 16-41% lower embodied CO_2 at no cost premium.
- Created an automated fracture-inspection pipeline: YOLO-seg instance segmentation, skeletonization with junction splitting, and Depth Anything V2 pixel-to-millimeter calibration quantify crack geometry; MobileSAM extracts specimen width for ranked safety assessment.
- Replaced hours of finite-element meshing with a physics-informed neural network (PINN) that learns stress-intensity factor K_I directly; matched handbook values within 0.9-2% and enabled microsecond edge-hardware scoring of K_{eff}/K_{IC} , including mixed-mode K_{II} .

Samsung Research - PRISM Program

Research Intern

Remote
Oct 2024 - Jun 2025

- Designed BQ2Viz, a multi-agent generative AI framework for answering business questions across one or multiple datasets with automated visualizations; combined ChromaDB semantic retrieval, analysis planning, and LangGraph chart generation.
- Improved accuracy to 84.57% on VisEval complex multi-table queries, surpassing the NVAGENT baseline, by adding a self-correction loop that executes generated chart code, diagnoses failures, and repairs outputs; reduced error rate to 11.5%.

Team Vyadh (Mars Rover Team), VIT

Autonomous Systems Engineer | [GitHub](#)

Vellore, India
Apr 2024 - Sep 2025

- Architected frontier-based autonomous exploration for unknown, GPS-denied terrain: clustered frontier centroids from live occupancy grids, selected the nearest reachable target, and dispatched Nav2 goals in a closed navigate-re-evaluate loop.
- Co-developed a C++/PCL perception server that classifies Intel RealSense D455 point clouds as pits or obstacles and uses odometry for trajectory recovery; fused RTAB-Map/OctoMap SLAM for 3D mapping and localization, contributing to 4th place at IRDC 2025 and 13th at IRC 2025.

PROJECTS

HazMap - Cross-Modal Hazard-Aware Autonomous Rover

[Preprint](#) | [GitHub](#)

ROS 2 Humble, Nav2, SLAM Toolbox, YOLOv8, TensorRT, CUDA, Python

- Built CRAFT, a cross-modal risk-aware exploration framework for GPS-denied inspection: sparse frontier traversal is augmented with CUSUM gas-anomaly detection, coarse-to-fine sweeps, and chemo-visual risk fusion to redirect navigation toward hazards.
- Integrated a TensorRT FP16-optimized YOLOv8 Nano perception pipeline with RGB-D map projection and costmap inflation; deployed at 24.3 ms inference latency on an NVIDIA Jetson Orin Nano alongside SLAM Toolbox, Nav2, and gas sensing.
- Validated in GADEN and physical deployments: detected high-concentration hazards within the first 2% of a mission, recorded mean gas-source localization errors of 0.54 m and 0.53 m in multi-source and maze environments, and delivered 0.845 mAP@0.5 visual-hazard detection.

Alpha Stack - Terminal-Native Coding Agent

[Preprint](#) | [GitHub](#)

Python, C++, NetworkX, Tree-sitter, Bubble Tea, Jinja2, Google GenAI, OpenAI SDK, OpenRouter, vLLM

- Built Alpha Stack, a terminal-native coding agent with a Go/Bubble Tea TUI communicating with a Python core through stdin JSON-RPC; converts prompts into persisted acceptance checklists, tool-augmented architecture documents, and implementation blueprints.
- Architected a complexity-adaptive, two-pass blueprint pipeline with deterministic structural validation and 20-worker parallel file generation; a C++/Tree-sitter DGAT resolves imports, tracks dependency blast radius, and injects exact module addresses instead of relying on model inference.
- Implemented a persistent planner/corrector fix loop with Docker-isolated build, test, and runtime verification; AlphaBench evaluates 40 tasks across CUDA, Go, Rust, and TypeScript with cost/failure telemetry, lifting project success from $\leq 45\%$ to 75% across four frontier models.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL

AI/ML: PyTorch, CatBoost, PINNs, CrewAI, LangGraph, LangChain, RAG, vLLM, YOLOv8, OpenCV

Robotics: ROS 2, Nav2, SLAM Toolbox, RTAB-Map, OctoMap, PCL, Intel RealSense SDK, NVIDIA Jetson, ESP32

Systems & DevOps: Docker, CUDA, GCP, Git, Linux, Tree-sitter, OpenRouter, Google GenAI, OpenAI SDK